

A Packing of 13 Four-Spaces in $\mathbb{R}^{10}$

Limin Ge*

October 9, 2026

Abstract

We give 13 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.599696. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.584278, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 13 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.599696 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 13$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.584278, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 13/12 = 2.600000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N13.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 62 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
129984810002390181358599221571341661143682127103294508325643988860579082
754385093947489718294125888966493014243450599635106836591852692737748640
198720923137995014771099052261203872078579337933
```

denominator:

```
500000000000821286565023286834111216327048317601798772310484998403151720
603106977682715908803788467327648670887673239114427208439012117485621314
675439249876763651435546339887195575085081673043
```

B Coordinates

13 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N13.txt`.

```
655753667673 -209665447973 -200431458597 -376181354299483002066 -404836918077 167519583713 264858806924255332058 -136923230391 -454754457220 -587096943124
121250158995 244112519603 -296027238337 -167541460421452911345 -360735196421 -321443110801 3162368736633841092525 90756932870 149460993816 198761063020
-323997001787 -365893799817 -351254036601 -187231374311201237680 -18743563063 7713945493 -295499631101-225912951682 -264269856805 58150027869 533550402728
51913074366 -14972586319 -127561541974 16930587288744194589292 670990251445 -45255862396 469315104959-187317825176 134408106741 250470831468 51685518922
151526439086 -252239567850 -430172868917 3241676626376484519 170948655137 234437313039 176783053537 -57004883391 61499743675 485491719071 -378402401959
311400872517 -646393680671 329811382544 1971929019058461114729 222369033309 -65726149454 25949429849350641210252 -155518991459 -76360282345 255503551820
-82622985319 253287136826 -130944044004 685500162706779604311 749543729378 -294037938175 -36845944764554366152681 -420261831358 -146261779433 69933658343
-276626258424 -117649315532 -217912127073 3970025801236699107492 -147193291622 463713083073 -537335746300254073825 353891171205 504073844919 274698703222
398724349962 301649271067 -435929849094 -8047049038682686230276 11196483223 -348867363173 -3253885142122025783097 -331218413623 463809258577 444787583954
292157465876 334528651615 429013546885 404392092991-562168128838 -170576798549 -323217237104 -77471933800407303128 -347037994355 -167813458849 -1030118482220
137710059076 -274331928098 140667211106 -2545124907320618591233 71638174020 -138170776781 350496975048-121389211817 195646757425 107419925526 -78777293615
-638787440211 326283815876 -125212933932 1394609965756709212525 -61500975054 289941127360 3523049980010216443608 -560223080350 402551189129 -457360623860
37056700370 273242070846 -345671790304 -566984140647180359535559 283275308576 173397260136 54318245256711143345367 453192173126 -428214003487 -424724716101
-552053962640 -523873332719 -80653583207 9381234350123576776737 333848524585 335909034288 -8505619715253003642806 -414451042662 -326485257747 -4836852311
-57866103206 -170939927232 2090189218 442967874773 -107479170619 159229002116 -292937650055 1009104169053599054305 -162445722408 -514762712759 200138343795
274192106556 276227487006 -323558869075 533588226163439832969313 401657212523 374784032748 12651378312772163504955 117989152926 -58275109931 -177361701978
-235087811880 9497686934 -381859623451 -269628902914706018653 -481143094413 380930443567 -2210985406636550316415 -164547249126 148878451361 273250143784
133840123816 4760882560 -647174674179 40938551280 -150764415594 387480667604 86871901214 -600765081722108619832 -191007714502 121650992566 -446184592706
16934524441 426643410570 381818081011 -165402382334-72293261560 150374275923 116536862006 3297318343029590641360 638821782603 315727305998 268774248832
-338900775738 426047898207 152999611908 10744490525253304415264 -399065879568 542484429546 54970241104896320754088 -53459750395 208541118635 -195369108576
-190499830371 -120847116642 20029254154 8249518416131089029165 265244860151 -18429408003 575473190597 -46744364954 -113574876387 308743878472 -585368502760
-123432453208 -126374962810 308225687348 -2165661726634553853775 9381846805 -252936594962 -1157552037118332049864 308426527613 -407681045828 -145813885200
-217211287541 521070662261 -261843294938 162685245308983214678 -47630403427 -49297469804 -1770362094103521704599 101111910534 227522411254 130291195179
499105396212 281889362820 -36890990552 158038914442541932220180 31990385959 287570572439 109652829892 16626645270 279653639 -353529273542 -407866485784
152317133836 619114627910 135048028658 13214427816976971459254 584104289378 493032639782 -17277822380255980580427 -432809125562 -310382381927 221919462910
-43283352453 375205846091 -156770413302 -76129910143029182992390 146905888093 383340906680 2463034804521671716179 -485150068669 90284925218 476345812337
-266140194894 -176617598311 -158536401809 -4102181238095534336 8982476897 550072450390 584154386545 369281774205 338002654030 -412960298633 -181725287632
244962837946 61924776803 354623135829 -393277260768-251299381107 376434240544 384707576293 -18740555955339834435471 -301516940625 387396005718 -339240764482
653796215010 -262604809922 -488555744499 -10814996663743912337 -154463958819 49480727230 129244253083-246483003951 -74384462796 -446310047319 165374104361
-262347975456 78663400426 -634255383242 97989731941-427062901709 668407228919 -340950266985 586448066231511967112 14362892752 405651004314 -286476865647
259604946561 335393633843 -501209318120 204180957972318650613537 -101408864060 426464189464 -1572043690578198292440 -399152273037 -163967784242 -486541982099
-19902215148 -861702085152 -152483899612 -237644961061730896085 486298671909 331444227290 -13098747017487257966929 443691150873 86713960768 215096461021
530932464684 -288112249197 83599855124 31274599498686924849925 168300938457 -11318843625 -399369248847313454905571 13969564302 -351185545090 131073868040
79905478149 -94044262011 -244811450841 -368618026474112399525287 -324864414734 169775037809 -553153944085206255105 161498529844 23353331782 232954703166
680166069124 42435503215 143406765304 -29744465185798837839668 18737847991 207946202317 158896375505 -79493070349 252845171884 224918633303 -379264547263
206360818717 190240328456 -234090224711 -20466457848256090271562 -81276580912 -258465993313 26628076663938447427985 -244380218168 726069046934 333456216146
-295153987595 49953186739 -25777603474 -5089264591238351465373 -401451668534 241464998115 -104837368932236304713 -499952979482 19747822775 294971359243
-176323247699 -40898229758 -169699406708 47195350339945606019758 -550264955141 32556638177 4792205512271500802162 119776901609 270145189929 52268707517
109685553341 41371523540 556364560081 -16065456496262168890616 -158248864911 -442408197734 24707836074243167769445 -207670603400 -445014618310 89774611709
-92128455358 96207546173 494215202839 357222258649 -258662015521 135031512724 -191972028981 -196594709320675507686 -227711808585 -42023500515 206028344692
389179619709 307980445014 181863084195 -12130656907241860987 448874610711 -161738264222 3417286562168832773856 -699002857822 14531923106 -526953621341
86909223391 137646089348 480261925474 -22931632387112057802777 -332511353445 253597221418 -851238225473867370628 -46759745231 -158035186752 505558802876
344494662406 250635418867 -671844508535 -455207757633750564361 -6018595541 -494193085611 388168982336
-453159549908 -10488367511 -303516712862 3862155495244465812023 -392040294629 -379761265715 -89133149063
```